State Estimation and Filtering for Robotic Perception

AA273

Independence, Conditional Independence,

Random Processes, Markovianity, Recursive Bayesian filtering



Stanford University



AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 1 due Fri Jan 16
- HW 2 out Fri Jan 16, due Fri Jan 23
- Waitlisted students: email me if you are still on waitlist and still interested in taking the class. Waitlist currently down to 16
- Project topic ideas posted
- Will soon post group signup sheet

Lecture 2:

- Independence, Conditional Independence, Graphical models
- Recursive Bayes,
- Random processes, Markovianity, Graphical models
- Recursive Bayesian filter

Recall: Joint distribution, independence, and conditioning

- Joint distribution of two random vectors *X* and *Y* is denoted as

$$p(x, y) = p(z) \qquad z = (x, y)$$

- *X* and *Y* are independent iff

$$p(x, y) = p(x)p(y)$$

- Conditional distributions

$$p(x \mid y) := \frac{p(x, y)}{p(y)}$$

Conditional probability

Recall: marginalization, factorization, law of total prob.

Factorization:

joint conditional marginal

$$p(x, y) = p(x \mid y)p(y)$$

(with conditioning) $$p(x, y \mid z) = p(x \mid y, z)p(y \mid z)$$

Marginalization:

$$p(x) = \sum_y p(x, y) \qquad p(x) = \int_y p(x, y)\, dy$$

(with conditioning) $$p(x \mid z) = \sum_y p(x, y \mid z) \qquad p(x \mid z) = \int_y p(x, y \mid z)\, dy$$

Law of total probability:

$$p(x) = \sum_y p(x, \mid y)p(y) \qquad p(x) = \int_y p(x, \mid y)p(y)\, dy$$

(with conditioning) $$p(x \mid z) = \sum_y p(x, \mid y, z)p(y \mid z) \qquad p(x \mid z) = \int_y p(x, \mid y, z)p(y \mid z)\, dy$$

Recall: Bayes' rule and probabilistic inference

- Assume *x* is a quantity we would like to infer from *y*
- Bayes rule allows us to do so through the inverse probability, which specifies the probability of data *y* assuming that *x* was the cause

Posterior / belief

Likelihood

Prior

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{\int_{x'} p(y \mid x')p(x')\,dx'}$$

Data

Normalizer

$$p(x \mid y) = \eta(y)p(y \mid x)p(x)$$

Const of proportionality, independent of x, Inverse of normalizer

(with conditioning)

$$p(x \mid y, z) = \frac{p(y \mid x, z)p(x \mid z)}{\int_{x'} p(y \mid x', z)p(x' \mid z)\,dx'}$$

Independence

Def (independent RVs): RVs x and y are independent iff $p(x, y) = p(x)p(y)$

Thm: The following are equivalent:

(i) $p(x, y) = p(x)p(y)$

(ii) $p(x \mid y) = p(x)$

(iii) $p(y \mid x) = p(y)$

$$p(x, y \mid z) = p(x \mid z)p(y \mid z), \qquad \text{equivalent to} \begin{cases} p(x \mid z) = p(x \mid z, y) \\ p(y \mid z) = p(y \mid z, x) \end{cases}$$

$$p(x, y \mid z) = p(x \mid z)p(y \mid z) \implies p(x, y) = p(x)p(y)$$

$$p(x, y) = p(x)p(y) \implies p(x, y \mid z) = p(x \mid z)p(y \mid z)$$

Conditional Independence

Def (conditionally independent RVs): RVs x and y are independent given z iff

$$p(x, y \mid z) = p(x \mid z)p(y \mid z)$$

Thm: The following are equivalent:

(i) $p(x, y \mid z) = p(x \mid z)p(y \mid z)$

(ii) $p(x \mid y, z) = p(x \mid z)$

(iii) $p(y \mid x, z) = p(y \mid z)$

Conditional independence does not imply independence!

$$p(x, y \mid z) = p(x \mid z)p(y \mid z) \not\Rightarrow p(x, y) = p(x)p(y)$$

E.g. Umbrellas and Puddles



$$Z = \begin{cases} 1 & \text{if its raining} \\ 0 & \text{if not raining} \end{cases}$$

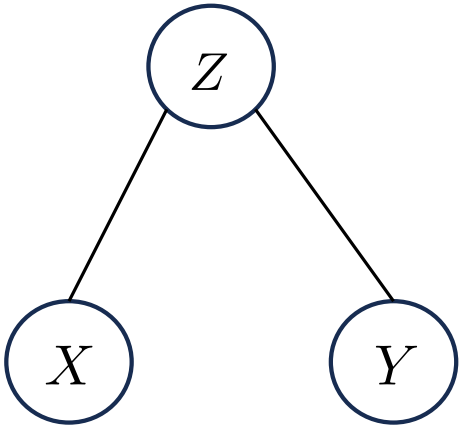




$$X = \begin{cases} 1 & \text{if see umbrellas} \\ 0 & \text{if no umbrellas} \end{cases}$$

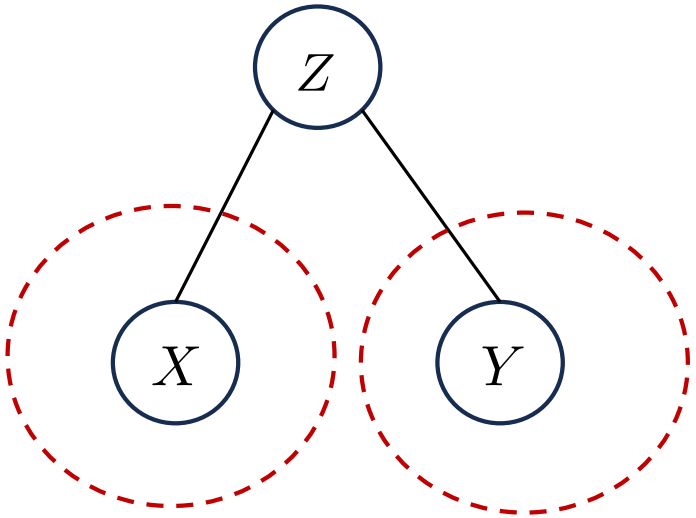
$$Y = \begin{cases} 1 & \text{if see puddles} \\ 0 & \text{if no puddles} \end{cases}$$

Graphical model:



Graphical models and cond. Indep.

- Cutting edge to Z Separates X from graph
- X cond. indep. of Y given Z
- Z is a “Markov blanket for” X

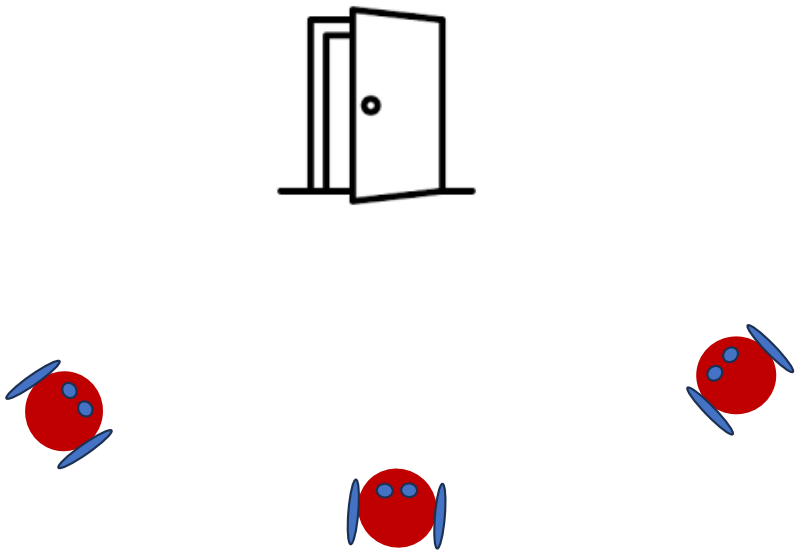


- Cutting edge to Z Separates Y from graph
- Y cond. indep. of X given Z
- Z is a “Markov blanket for” Y

$$p(x \mid y, z) = p(x \mid z)$$

$$p(y \mid x, z) = p(y \mid z)$$

Cond. Indep. Meas. Robotics Example



Conditionally Independent Measurements

Define shorthand:

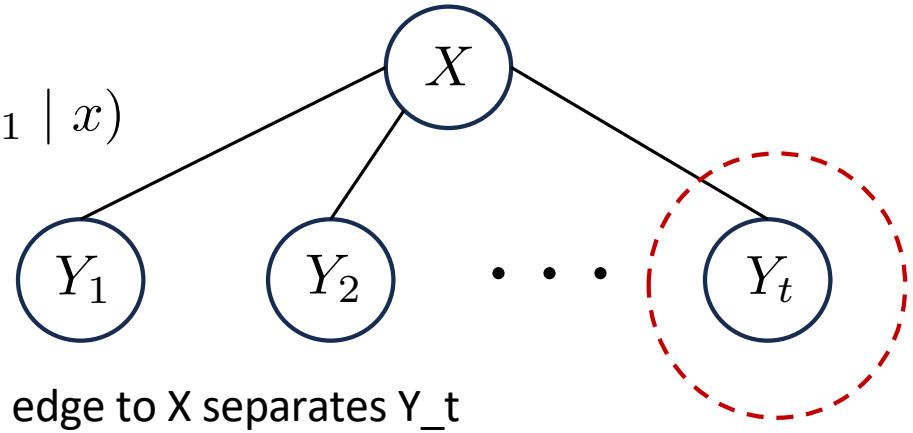
$$y_{1:t} := (y_1, y_2, \ldots, y_t)$$

Factor the likelihood:

$$p(y_{1:t} \mid x) = p(y_t \mid y_{1:t-1}, x) p(y_{1:t-1} \mid x)$$

$$= p(y_t \mid x)$$

$$p(y_{1:t} \mid x) = p(y_t \mid x) p(y_{1:t-1} \mid x)$$



Cutting edge to X separates Y_t from graph:

$$p(y_t \mid y_{1:t-1}, x) = p(y_t \mid x)$$

Simplify:

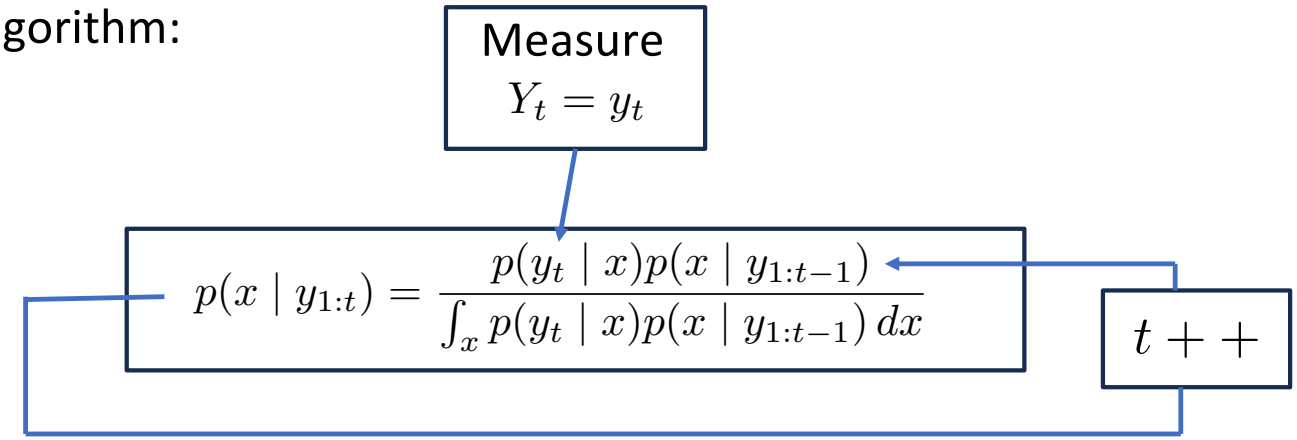
$$p(y_{1:t} \mid x) = \prod_{\tau} p(y_\tau \mid x)$$

Sequential Bayes' Rule

$$p(x \mid y_{1:t}) = \frac{p(y_t \mid x)p(x \mid y_{1:t-1})}{\int_x p(y_t \mid x)p(x \mid y_{1:t-1})\, dx}$$

(Derivation in notes / on board)

Recursive algorithm:



Random Processes

Collection of jointly distributed, time indexed RVs:

$$X_{1:t} = (X_1, X_2, \ldots, X_t)$$
$$X_{1:t} \sim p(x_{1:t})$$

Usually defined by special temporal correlation, or lack of temporal correlation.

E.g., white noise process (no temporal correlation)

$$p(x_{1:t}) = \prod_{\tau} p(x_\tau)$$

E.g., stationary white noise process (a.k.a, independent and identically distributed)

$$p(x_{1:t}) = \prod_{\tau} p(x_\tau) \qquad p(x_t) = p(x_\tau) \quad \forall\, t, \tau \qquad\qquad X_{1:t} \quad \text{i.i.d.}$$

Random Processes

E.g. Gaussian white noise process

$$p(x_{1:t}) = \prod_{\tau} \mathcal{N}(\mu, \Sigma)$$

Stationary Gaussian white noise process

$$p(x_{1:t}) = \prod_{\tau} \mathcal{N}(\mu_\tau, \Sigma_\tau)$$

Non-stationary/time varying Gaussian white noise process

Illustration on board / notes